GODAVARI PUSHKARALU 2027

DIGITAL PILGRIM EXPERIENCE

Project Proposal and Prototype Documentation

Selected Hackathon Track: Problem Statement 12 – Digital Pilgrim Experience and Heritage Access

Accessible event information · Multilingual digital guides · Heritage access · Virtual participation

Prepared as a concept and prototype document. All event content in the demo is illustrative until verified by authorised organisers.

1. Executive summary

This proposal presents a digital pilgrim experience for Godavari Pushkaralu 2027, focused exclusively on Problem Statement 12. The platform is intended to help pilgrims obtain event-related information and engage with cultural or heritage content through virtual participation, digital guides and suitable technology-enabled services.

The proposed solution is a mobile-friendly web platform with multilingual information, searchable visitor guidance, a digital FAQ guide, heritage learning content and a dedicated area for organiser-approved virtual content. The current prototype demonstrates the core interface and a local rule-based guide. It does not claim to provide live official feeds or production integrations.

2. Source problem statement and scope

The uploaded Godavari Pushkaralu 2027 concept note states: “Develop an accessible digital experience that helps pilgrims obtain event-related information and engage with relevant cultural or heritage content through virtual participation, digital guides or other suitable technology-enabled services.”

This project addresses only Problem Statement 12. It does not implement the other eleven challenge tracks. The cross-cutting requirements are treated as design and validation requirements: high-footfall suitability, field usability, multilingual accessibility, reliability under local conditions, privacy and cybersecurity, interoperability, affordability, scalability and post-event use.

3. Users and needs

Primary users include pilgrims and visitors seeking verified event information; people who prefer regional languages; elderly visitors and people who benefit from accessible content; people participating remotely; and field personnel who may need clear, current information to guide visitors.

Key needs include finding official event guidance, understanding local heritage, accessing content in a preferred language, using the service on common mobile devices and distinguishing verified information from illustrative or outdated material.

4. Proposed modules

Event information: organiser-approved schedules, announcements, visitor instructions and verified route/facility guidance. Time-sensitive information should show source and last-updated time.

Multilingual digital assistance: English, Telugu, Hindi and Tamil interface labels in the prototype, with a pathway to professionally reviewed translations, voice input and text-to-speech.

Digital guide: a FAQ-based guide for common topics. A production version may use retrieval-augmented generation grounded only in approved content and should cite source pages.

Heritage access: reviewed articles and digital exhibits about the Godavari region, traditions, places, architecture and stories. Content should be checked by qualified local historians and relevant cultural representatives.

Virtual participation: a dedicated section for authorised livestreams, recorded cultural sessions, audio guides and virtual exhibits, with captions and transcripts where available.

Accessibility: responsive design, readable text, keyboard navigation, screen-reader testing, captions, audio descriptions and low-bandwidth support.

5. Prototype implementation

The prototype uses Python Flask for the backend and HTML, CSS and JavaScript for the user interface. A JSON endpoint provides basic topic-based answers. It includes a language selector, event information panel, searchable visitor-information cards, heritage cards, virtual participation placeholder and a larger-text control.

The digital guide is intentionally rule-based so the prototype can run without paid AI credentials or external model access. It should not be represented as a production AI assistant.

6. Technical architecture

Presentation layer: responsive web interface and language selector.

Application layer: Flask routes serving pages and a JSON guide endpoint.

Content layer: organiser-approved event information and expert-reviewed heritage content.

Future integration layer: documented APIs for approved event schedules, maps, announcements and official media.

Administration layer: a future secure content-management system with role-based access, audit logs and review/approval workflow.

7. Privacy, cybersecurity and governance

The prototype does not require accounts and does not intentionally collect visitor personal information. A production deployment should collect only data necessary for the service, provide clear privacy notices, define retention periods and apply appropriate access controls.

Use HTTPS, secure authentication for administrators, role-based permissions, input validation, protected secrets, backups, audit logging, vulnerability testing and an incident-response process. Document data ownership, software/IP rights, licensing, hosting, access, retention, backups and exit/handover arrangements. Follow applicable data-protection, cybersecurity and government information-security requirements.

8. Reliability, affordability, interoperability and scalability

Design pages to remain lightweight under variable local network conditions. Cache approved essential information where appropriate, show timestamps and provide clear fallback guidance when live sources are unavailable.

Prefer open standards and documented APIs where feasible. Keep content structures configurable so additional locations and languages can be added without rebuilding the application. Assess hosting, translation, support and maintenance costs before a pilot; scale infrastructure based on measured demand.

9. Post-event use

After the event, the platform can continue as a digital heritage and visitor-information resource, subject to content rights and approval. Future use may include regional heritage articles, recorded cultural sessions, accessible learning materials and information for later public events. Assign owners for content review, updates, hosting,

moderation and maintenance.

10. Implementation roadmap

Phase 1 – Discovery: confirm users, official content owners, source systems, languages, accessibility needs and approved content.

Phase 2 – Core experience: refine event information, language interface, search, heritage pages and digital guide.

Phase 3 – Authorised integrations: connect official schedules, route/map data and virtual media where access is approved.

Phase 4 – Assurance: conduct accessibility, privacy, cybersecurity, low-bandwidth and load testing.

Phase 5 – Pilot: test with representative pilgrims, visitors and field personnel; gather evidence and fix issues.

Phase 6 – Handover: document hosting, support, maintenance, content ownership and post-event reuse.

11. Evaluation measures

Content accuracy and freshness; user success in finding information; language comprehension; accessibility task completion; page load and uptime under expected traffic; guide answer correctness against the approved knowledge base; user satisfaction; support burden; and post-event content reuse.

Specific performance targets should be agreed with the responsible authority before pilot implementation. Do not invent targets without baseline information.

12. Risks and mitigations

Outdated information: named content owners, review workflow and update timestamps.

Incorrect translation: professional review and approved terminology.

Unverified heritage content: expert and community review.

Poor connectivity: lightweight pages and cached essential information.

Unauthorised media or copyright issues: rights checks and organiser approval.

Cybersecurity incidents: secure development, access controls, testing, backups and response procedures.

13. Current prototype limitations

All event details and visitor guidance in the prototype are sample content. No live schedule, official announcement, map, booking or government-system feed is connected.

The guide uses a local FAQ mapping, not a live AI model. The virtual participation area is a placeholder and does not stream video.

A production-ready service would require approved data, integrations, content management, hosting, accessibility audits, security review, load testing and operational support.

14. Conclusion

The proposed Digital Pilgrim Experience is aligned to Problem Statement 12 only. It combines accessible event information, multilingual digital assistance, heritage learning and virtual participation. The prototype provides a demonstrable starting point while making clear which integrations and governance safeguards remain necessary before any public pilot.

Appendix A – Running the prototype

1. Install Python 3.9 or later.
2. Extract the source-code ZIP and open the project folder in VS Code.
3. Create a virtual environment: `python -m venv .venv`
4. Activate it on Windows: `.venv\\Scripts\\activate`
5. Install Flask: `pip install -r requirements.txt`
6. Start the app: `python app.py`
7. Open `http://127.0.0.1:5000` in a browser.
8. Check `http://127.0.0.1:5000/health` for the local health response.

Appendix B – Files included

app.py; templates/index.html; static/style.css; static/app.js; requirements.txt; README.md; project presentation PowerPoint.